

Bachelor of Technology — Engineering Physics

Table 1: Engineering Physics — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	36
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	37
	Discipline Electives	29
Electives	Free Electives	25
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Engineering Physics — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EP-301	Engineering Mathematics 2	3-1-0-4	4
2	EP-321	Foundations of Electrodynamics	3-0-0-3	3
3	PH-301	Quantum Mechanics and Applications	2.5-0.5-0-3	3
4	EE-223P	Reverse Engineering	1-0-0-1	1
5	EP-403	Physics of Atoms and Molecules	3-0-0-3	3
6	PH-302	Introduction to Statistical Mechanics	2.5-0.5-0-3	3
7	PH-501	Solid State Physics	3	3
8	EE-311	Device Electronics for Integrated Circuits	3-0-0-3	3
9	EP-302	Computational Methods for Engineering	2-0-2-3	3
10	EP-402P	Engineering Physics Practicum	1-0-5-4	4
11	EP-401P	Engineering of Instrumentation	1-0-5-4	4
12	PH-502	Photonics	3-0-0-3	3
Total				37

Table 3: B. Tech Engineering Physics — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS	IC-181	Indian Knowledge Systems	2-0-2-3	3
10	(choose 1)	IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Engineering Physics — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Engineering Physics — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EP-301	Engineering Mathematics 2	3-1-0-4	4
4	DC	EP-321	Foundations of Electrodynamics	3-0-0-3	3
5	DC	PH-301	Quantum Mechanics and Applications	2.5-0.5-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Engineering Physics — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-223P	Reverse Engineering	1-0-0-1	1
2	DC	EP-403	Physics of Atoms and Molecules	3-0-0-3	3
3	DC	PH-302	Introduction to Statistical Mechanics	2.5-0.5-0-3	3
4	DC	PH-501	Solid State Physics	3	3
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Engineering Physics — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-311	Device Electronics for Integrated Circuits	3-0-0-3	3
2	DC	EP-302	Computational Methods for Engineering	2-0-2-3	3
3	DC	EP-402P	Engineering Physics Practicum	1-0-5-4	4
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Engineering Physics — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EP-401P	Engineering of Instrumentation	1-0-5-4	4
2	DC	PH-502	Photonics	3-0-0-3	3
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Engineering Physics — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	EP-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Engineering Physics — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	EP-010	Internship		2
2	MTP	EP-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 29 credits of **Discipline Electives** and 25 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Engineering Physics — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-208	Mathematical Foundations of Computer Science	4
2	CS-241	Introduction to Cryptography	3
3	CS-304	Formal Languages and Automata Theory	3
4	CS-309	Information and Database Systems	4
5	CS-511	Applied Probability	2
6	CS-671	Deep Learning and Applications	4
7	CS-672	Advanced Topics in Deep Learning	4
8	DS-201	Data handling and visualization	3
9	DS-301	Mathematical Foundations of Data Science	4

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
10	DS-401	Optimization for data science	3
11	DS-404	Information Security and Privacy	3
12	DS-416	Fundamentals of Machine Learning (CS-669)	4
13	EE-203	Network Theory	3
14	EE-211	Analog Circuit Design	3
15	EE-307	Theory of Measurement	3
16	EE-511	Computer Vision	4
17	EE-512	CMOS Analog IC Design	4
18	EE-519P	CMOS Digital IC Design Practicum	2
19	EE-520	Microelectronic Devices and Modelling	3
20	EE-524	Digital MOS LSI Circuits	3
21	EE-534	Probability and Random Processes	3
22	EE-551	Applied Photonics for Scientists and Engineers	3
23	EE-593	Low Power VLSI Circuits	3
24	EE-611	VLSI Technology	3
25	EE-614	Optical Communication Systems	3
26	EE-621	Radiating systems	4
27	EN-502	Emerging Energy Sources	3
28	EP-201	Reverse Engineering	1
29	EP-401	Engineering of Instrumentation	3
30	EP-402	Engineering Physics Practicum	3
31	EP-502	Informatics for Materials Design	3
32	IC-253	Programming and Data Structures	3
33	MA-210	Real and Complex Analysis	3
34	MA-511	Real Analysis	4
35	MA-512	Linear Algebra	4
36	MA-513	Ordinary Differential Equation	4
37	MA-516	Topology	4
38	MA-521	Topology and Functional Analysis	4
39	MA-522	Partial Differential Equations	4
40	MA-560	Nonlinear Dynamics and Chaos	3
41	ME-210	Fluid Mechanics	3
42	ME-303	Heat Transfer	3
43	ME-307	Energy Conversion Devices	3
44	ME-308	Manufacturing Engineering	3
45	ME-503	Micro/Nano Scale heat transfer	3
46	ME-509	Nano-manufacturing	3
47	ME-510	Advanced Manufacturing Processes	3
48	ME-603	Advanced Fluid Mechanics	3
49	ME-615	Applied Computational Fluid Dynamics	3

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	PH-503	Laser and Applications	3
51	PH-504	Organic Optoelectronics	3
52	PH-507	X-rays as a Probe to Study Material Properties	3
53	PH-508	Magnetism and Magnetic Materials	3
54	PH-511	Mathematical Physics	4
55	PH-512	Classical Mechanics	4
56	PH-513	Quantum Mechanics	3
57	PH-514	Electronics	3
58	PH-515P	Physics Laboratory	3
59	PH-516	Research Project I	2
60	PH-521	Electromagnetic Theory	4
61	PH-525P	Electronics Lab	3
62	PH-528	Introduction to General Relativity	3
63	PH-530	Cosmology-I	3
64	PH-550	Introduction to Quantum Optics	3
65	PH-579	Quantum Computation and Information	3
66	PH-600	Research Methodology	1
67	PH-601	Mesoscopic Physics and Quantum Transport	3
68	PH-603	Advanced Condensed Matter Physics	3
69	PH-604	Optical Properties of Solids	3
70	PH-605	Superconductivity	3
71	PH-606	Quantum Field Theory	3
72	PH-607	Physics of Ultracold Quantum Gases	3
73	PH-608	Computer assisted quantum mechanics	3
74	PH-609	Theory of quantum collision and spectroscopy	3
75	PH-612	Nuclear and Particle Physics	3
76	PH-613	Special Topics in Quantum Mechanics	3
77	PH-615P	Mini Thesis I	3
78	PH-621	Computational Methods for Physicists	4
79	PH-625	Data Analysis in Particle Physics	4
80	PH-626	Elementary Theoretical Particle Physics	3
81	PH-627	Topological Quantum Matter	3
82	PH-701	Introduction to Molecular Simulations	4
83	PH-706	Introduction to Stochastic Problems in Physics	3
84	QS-501P	Experiments in Quantum Optics	3
85	QT-301	Survey of Quantum Technologies and Applications	3
86	QT-406	Introduction to Quantum communication	3
87	QT-407	Introduction to Quantum Sensing	3
88	QT-509	Engineering Foundations of Quantum Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
89	EP-498P	Major Technical Project - I (compulsory; counted as DE)	4
90	EP-499P	Major Technical Project - II (compulsory; counted as DE)	4